Fitted-Model Results

lab02_espenilla01 • Sum of Two Logistics • 27 August 2026

Model: $h = c + a_1 \cdot S_1(t) + a_2 \cdot S_2(t)$ $n = 288$ $p = 7$ dof = 281 Levenberg–Marquardt fit on the raw log.

1. Fitted Parameters

Parameter	Estimate	SE	t	p
c	14.3493	0.0097	1483.2	<0.001
a_1	8.9160	0.7532	11.84	<0.001
k_1	0.5117	0.0213	24.05	<0.001
t_1	31.5198	0.0906	348.0	<0.001
a_2	−3.7212	0.7594	−4.90	1.6×10^{-6}
k_2	0.2831	0.0138	20.57	<0.001
t_2	37.5461	1.0577	35.50	<0.001

t = estimate / SE; p is a two-tailed t on 281 df. The two inflections occur at approximately t_1 = 31.5 h and t_2 = 37.5 h.

2. Goodness of Fit and Area

SSE	R^2	s	Runs	A (quad)	Abs. error	Trapezoid	Gap
1.8017 m²	0.999114	0.0801 m	17	1260.9708 m·h	5.1×10^{-7}	1260.9963 m·h	−0.002%

SSE = Σ residual²; $R^2 = 1 - SSE/SST$; $s = \sqrt{SSE/dof}$. Area is in metre-hours (m·h), representing level × time, not volume. Quadrature integrates the fitted curve; trapezoid joins the 288 raw readings.

3. Residual Diagnostics

- Overall: R^2 = 0.999 is very high, but the residuals are not a clean random cloud; they drift, form same-sign stretches, and widen where the level changes fastest.
- Local centring: residuals move from +0.0073 m during the first calm period to −0.0303 m in the tail, indicating the model cannot fully follow the slow drawdown.
- Runs: 17 runs versus approximately 144 expected; the longest stretch is 54 readings (≈4.3 h), indicating a shape mismatch rather than random noise.
- Variance: residual spread is approximately 0.8× wider across the limb, suggesting variance tracks dh/dt.
- Largest error: 22 cm, substantially larger than the 1 cm logger resolution, indicating genuine model misfit.

Conclusion: The two-logistic model gives an excellent numerical fit by R^2, but structured residuals show that it does not fully capture the observed slow drawdown.